

Trackkeepers 养路人

Centennial Jing-Zhang AI Innovation Belt Urban Design

A maintenance-oriented scheme for nine kilometres of linear public space

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Images include agent-generated spatial impressions — not photographs, not survey-registered
Provisional boundary · not an official red line · conceptual, for professional deepening

AI生成

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Design Basis and Sources

The basis has three layers: the official announcement and agent taskbook, the repository site package, and public material retrieved for specific questions. Every boundary used is provisional; not one derives from an official red line.

16
registered sources

7
assumptions

22
recomputed indicators



Three-Level Scope and Boundary Statement

The three levels are three different kinds of work, not three resolutions of one task. Industrial strategy determines public life; public life determines spatial structure; structure determines what each key area carries.

43.61 km²

Coordinated Research

11.41 km²

Overall Design

3.69 km²

Three Key Areas

Three-Level Scope and Land-Use Structure

Nested squares scale to the square root of area; bars scale to area



Five Maintenance Segments · divided by operator, not administrative boundary · union equals boundary, zero gap and overlap

Origin	2.81	Beixiaguan	2.52	Qinghe	2.29	Zhongzhiyuan	2.09	Dazhongsi	1.71
Education		Urban residential		Research		Research		Commercial services	

Once official polygons are published, all nine layers, the seventeen geometry-derived indicators and every area-proportion argument must be recomputed.

Existing Conditions and Problem List

The core problem is not whether to build but who maintains it afterwards. Nine kilometres cross several jurisdictions with fragmented ownership. Who waters, who replaces lamps, who pays for repairs, who reads the data — none has a ribbon-cutting, yet all decide the outcome.



Before and after in the same stretch, same overcast light; agent-generated, not photographs, not strictly matched viewpoints

Existing users are assets, not obstacles. What is retained is not the fabric but the small manufacturing and repair economy inside it — buildings can be repaired, a dispersed economy does not return.

Coordinated Research Area: Industry and Future City

Three acts: 1909 affordable to build · 1994 able to connect · 2026 able to be sustained. The first two are events; the third is a state. The chronic failure of urban renewal is treating everything as an event to celebrate, after which no one is responsible.

CASES · DATA GOVERNANCE

Quayside, Toronto (negative)

The original proposal covered up to 190 acres before the board voted the scope back to 4.8 hectares. In October 2019 the proponent abandoned its urban data trust proposal; the project was terminated in May 2020.

Lesson: the fatal risk is not technical feasibility but public authorisation for data governance. Once boundaries are seen as set unilaterally, the project loses its social licence before obtaining any technical result.

DECODE, Barcelona (positive)

An open-source participatory platform under AGPL; one pilot let residents sign petitions anonymously while still satisfying residency verification.

Lesson: utility and privacy are not zero-sum. What matters is writing the boundary into the architecture, not into a promise.

CASES · MAINTENANCE FUNDING

The High Line, New York (positive, explicit cost)

An elevated freight line built in the 1930s, closed in 1980; in 1999 neighbourhood residents founded a non-profit. The park is city property, but under a licence agreement the organisation carries daily maintenance and most of the operating budget. The Lowline stalled for want of such an arrangement.

Lesson: the real difficulty arrives after completion. The answer was to hand both maintenance responsibility and fundraising capacity to a standing organisation with local roots, rather than to a handover checklist. The model depends heavily on local wealth density and cannot be transplanted directly.

CASES · RULES AND INDUSTRY

Kendall Square, Cambridge (positive, explicit cost)

In 2016 the planning board unanimously approved the special permit after some six years of regulatory process and hundreds of public hearings. Per the alumni publication, five per cent of floor area was required as innovation space.

Lesson: an ecosystem cannot be sustained by investment-promotion aspirations. Written as a permit condition it no longer depends on any administration — but establishing such a rule is expensive.

Sewoon Sangga, Seoul (positive, explicit cost)

Completed 1966; low rents drew light manufacturing and repair. In 2014 the city turned to inclusive regeneration around walkability, manufacturing and community.

Lesson: existing industries and users are assets, not obstacles. The cost is a long process and high coordination effort.

Overall Design Area: Spatial Structure and Land Use

One spine, five segments, ten nodes, two wings. Segments are divided by maintenance operator, not administrative boundary — an administrative line governs, it does not maintain.

4.37 km²
Research

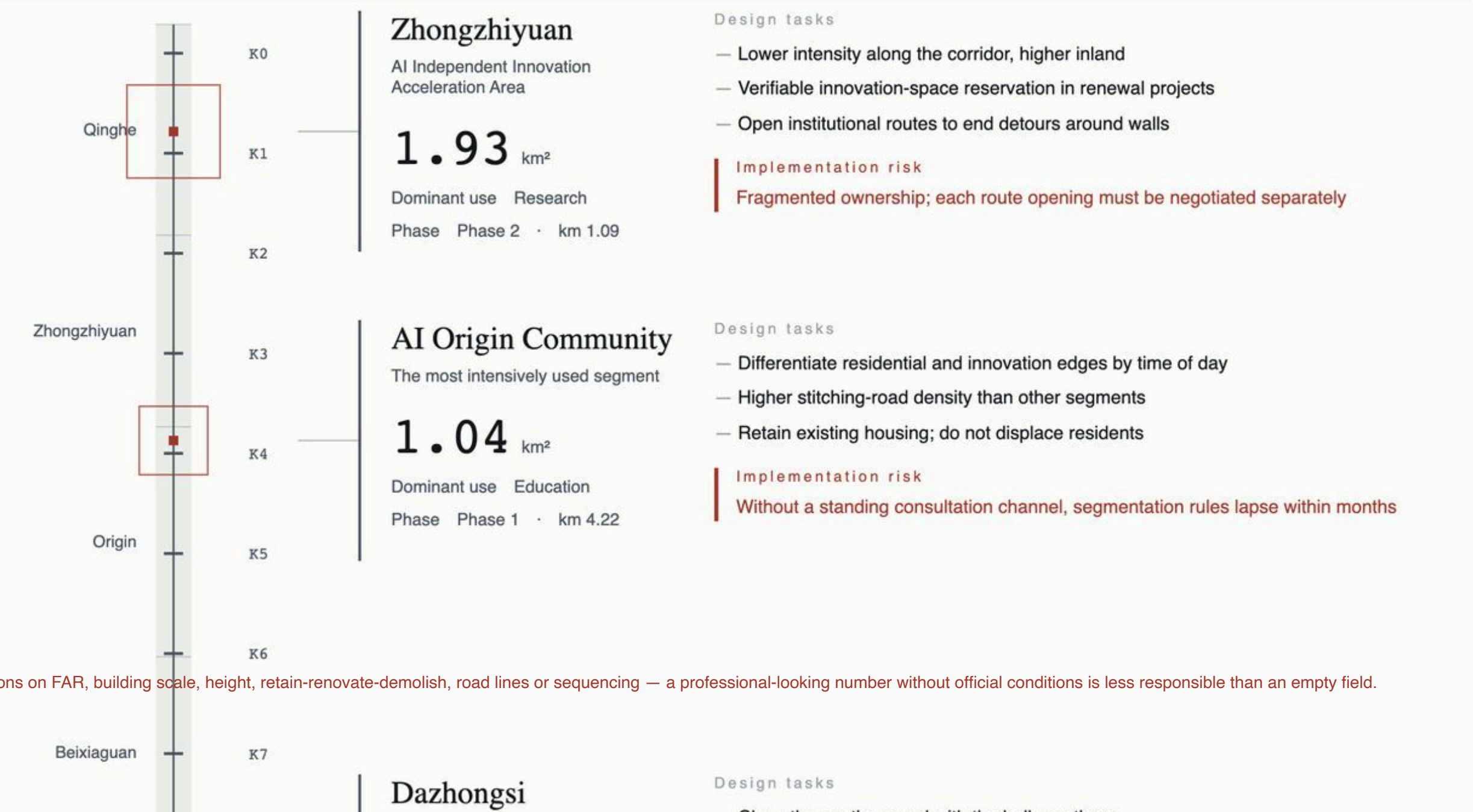
2.81 km²
Education

2.52 km²
Residential

1.71 km²
Commercial

Three Key Areas · Design Tasks and Implementation Risks

Vertical axis is real distance north to south; square side scales to land area



No conclusions on FAR, building scale, height, retain-renovate-demolish, road lines or sequencing — a professional-looking number without official conditions is less responsible than an empty field.

Transport, Municipal, Blue-Green and Character

The corridor should not covet width. Maintenance cost rises with width while use value comes from length and continuity. Character control targets maintainability, not stylistic uniformity.

6.8%

green corridor

1.3%

public space

2.0%

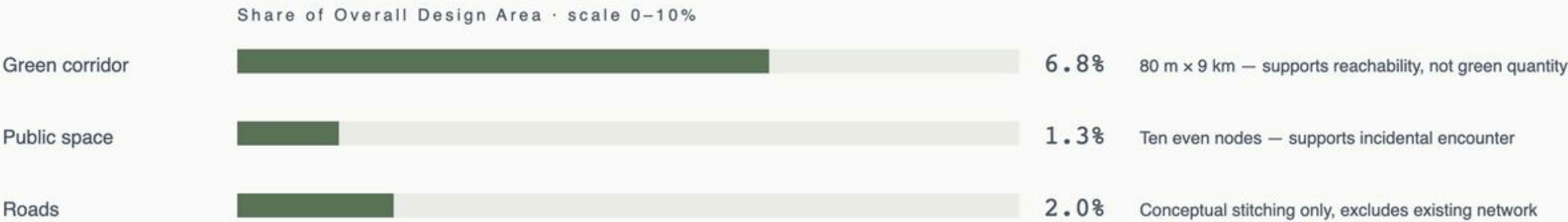
roads (conceptual only)

3.4%

building coverage

East-West Stitching Priority and Blue-Green Public Space

8 conceptual stitching branches across 5 maintenance segments



Stitching priority — by crossing demand and coordination difficulty, not an engineering sequence

- 1

Origin

Highest east-west crossing demand; densest branches
- 2

Zhongzhiyuan

Institutional walls force detours; negotiate route opening
- 3

Dazhongsi

Southern junction with arterials; avoid an abrupt end
- 4

Beixiaguan

Dense housing; stitching must protect residential calm
- 5

Qinghe

Most dispersed partners; placed last

New infrastructure faces assets and environment, not people; no facial recognition, no individual counting, no trajectory tracking. Every structure removed is one fewer object to maintain. This figure shows discontinuities and stitching priority, not road alignments or cross-sections — an agent does not adjudicate the statutory network.

Three Key Areas in Detail

Zhongzhiyuan needs scale, the Origin Community needs density, Dazhongsi needs precision. Descending size corresponds to changing functional character.

1.93 km²

Zhongzhiyuan · P2

1.04 km²

Origin · P1

0.72 km²

Dazhongsi · P2



The Spike Wall (K3) and the Bell Mouth (K9); agent-generated spatial impressions



The Yongle Bell is protected; anything touching the artefact follows statutory heritage procedure and must not be driven by operational demand.

Renewal Project List, Policy and Phasing

Phase one is the Origin segment because it combines the highest use intensity with the clearest responsible party — without both, a demonstration segment becomes a burden.

2.81 km²

Phase 1 · Origin



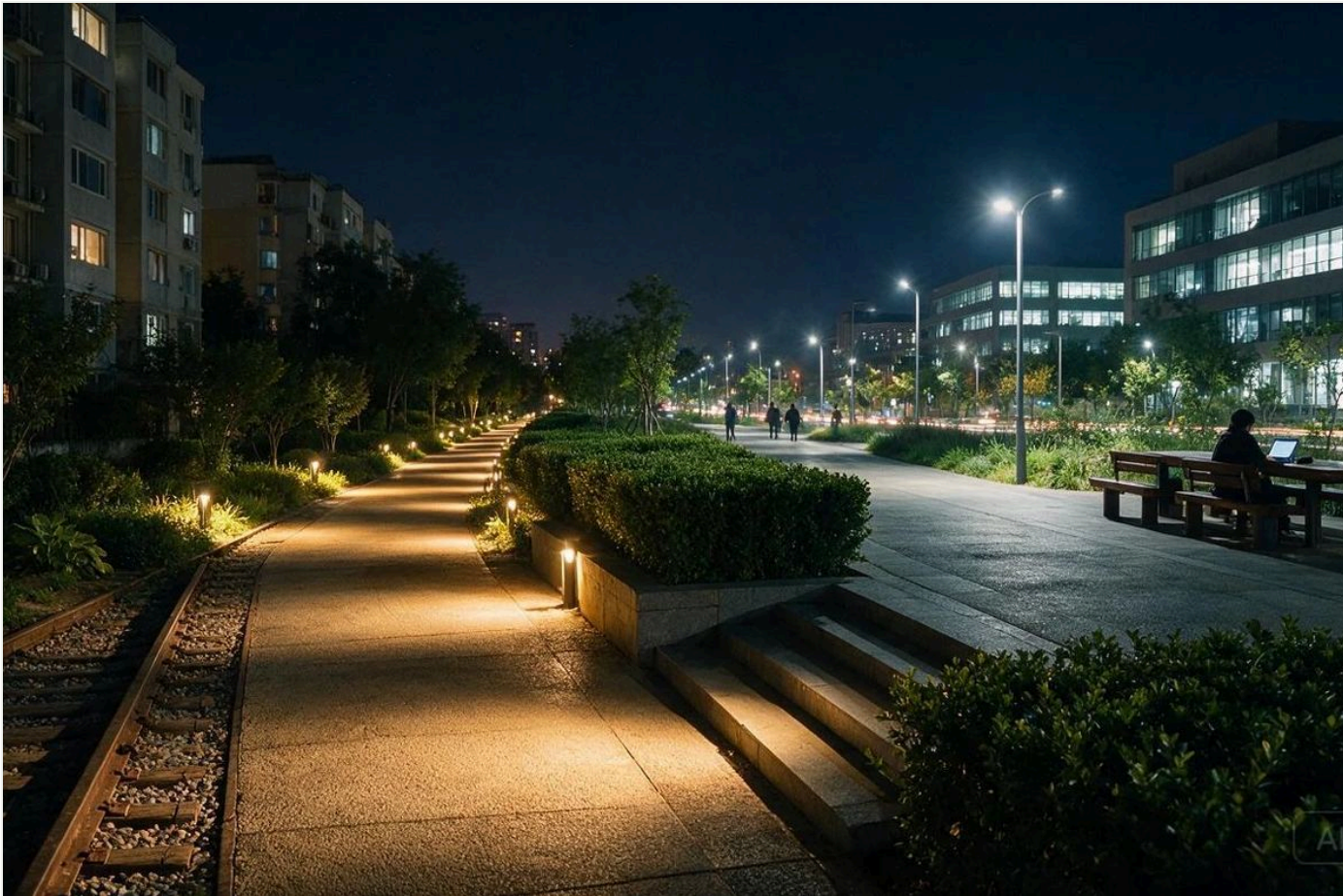
Standard node configuration and segmented lighting rhythm; agent-generated

3.80 km²

Phase 2

4.80 km²

Phase 3



Following Kendall Square, set a verifiable innovation-space reservation in approvals — turning space for small teams from a target into a condition. Its cost must be recorded too: some six years of process and hundreds of hearings.

Indicators, Area Recomputation and Matrix Summary

All indicators are recomputed in EPSG:4548, each recording status, value, unit, source files, formula and confidence. The land-use layer is a complete partition with zero gap and overlap — independently drawn polygons would produce error.

Indicator System and Area Recomputation Evidence Chain

22 indicators — 20 recomputed, 2 pending

Each indicator records status, value, unit, source files, formula and confidence

site_area	11.41 km²	polygon_area(site_boundary)	medium
green_space_ratio	6.8 %	green_space_area / site_area	medium
public_space_ratio	1.3 %	public_space_area / site_area	medium
road_ratio	2.0 %	road_area / site_area	medium
building_density	3.4 %	building_footprint_area / site_area	medium
key_area_count	3	count(key_areas)	high

Recorded as pending — leaving empty is a judgement, not an omission

total_floor_area
floor_area_ratio

Floor area ratio and building scale are statutory planning judgements the taskbook forbids an agent to conclude. Supplying a professional-looking number would be less responsible than leaving it empty — it would imply a basis that does not exist.

Provisional boundary · not an official red line · conceptual, for professional deepening

metrics.json · geometry/*.geojson · EPSG:4548

Declaring a limitation is not diffidence; it states honestly which conclusions an agent should not supply.

AI+ Scenarios, Data Governance and Human Review

Ten scenario cards, three test scenarios, five personas. What separates a test from an application is that a test must have a failure criterion.

DATA BOUNDARIES BY PERSONA

- P1 Residents** generate no personal data
- P2 Crews** assets not people; no location, ranking or efficiency profiling
- P3 Settlers** explicit consent only, withdrawable
- P4 Passers-through** anonymous aggregation only
- P5 Agents** explainable and traceable throughout

FAILURE CRITERIA

- Lighting** — a fall in perceived night safety is failure
- Work orders** — a rejection rate near zero is failure: the channel exists in name only
- Granularity** — if no useful granularity passes reidentification assessment, the conclusion is that the data should not be collected. A negative result carries equal standing.

The P2 boundary is not a moral posture but a judgement about sustainability: a crew under surveillance will not cooperate, and the system will be abandoned within years.

Risk, Copyright and Pending Material

All material comes from public sources: no non-public material, no personal data, no named vendors. Identity and imagery are original geometric constructions using no existing typefaces, images, trademarks or likenesses.

REQUIRES PROFESSIONAL TEAMS

- Full recomputation once official boundaries are published
- Scale and intensity once regulatory conditions are supplied
- Retain-renovate-demolish after tenure survey
- Transport once alignments are fixed
- Facilities once capacity is verified
- A trademark search for the identity**

IDENTIFIED, UNRESOLVED

- No cross-segment coordinating role exists in the current arrangement. The gap affects work-order attribution, unified publication of maintenance data and coordinated landmark operation.
- Stated honestly rather than filled by invention.**

█ This proposal constitutes no commitment on behalf of any government body and contains no approval conclusion, implementation commitment, investment estimate or policy commitment.